



School of Data Science

DASC 605
Dr Trent
Buskirk
Chapter 4
Lab

For this lab go to: <https://artofstat.com/web-apps>

- Scroll down about 2/3 of the way to see this set of apps:

Confidence Intervals & Significance Tests (One Sample)

Inference for a Proportion

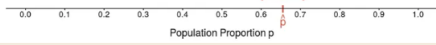
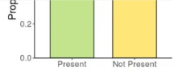
Estimate of Population Proportion:

Point Estimate	Standard Error	Margin of Error
0.6551	0.0287	± 0.0524

Confidence Interval:

Population Parameter	Lower Bound	Upper Bound	Confidence Level
Proportion p	0.6027	0.7075	95%

95% Confidence Interval
[0.6027, 0.7075]

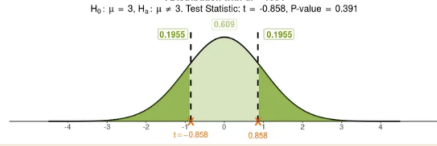


Inference for a Mean

Hypothesis Test:

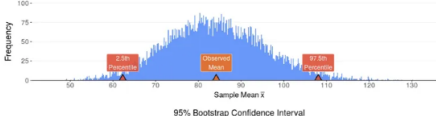
Population Parameter	Null Hypothesis	Alternative Hypothesis	Test Statistic t	P-value
Mean μ	$\mu = 3$	$\mu \neq 3$	-0.8581	0.3910

t Distribution with df = 1554
 $H_0: \mu = 3, H_a: \mu \neq 3$. Test Statistic: $t = -0.858$, P-value = 0.391



Bootstrap for One Sample

Bootstrap Samples	Statistic	Unique Values	Mean	Std. Dev.	Min	Q1	Median	Q3	Max
11,000	Mean	3,515	84	11.7	49.1	75.9	83.6	91.8	130



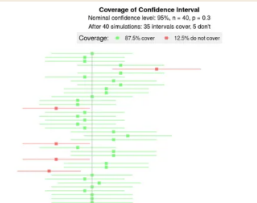
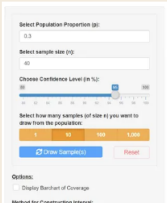
95% Bootstrap Confidence Interval
[82.3, 108.1]

Find **confidence intervals** or test **hypotheses** about a population *proportion*. Obtain the margin of error or the z-test statistic and visualize the interval or the **P-value** on a graph.

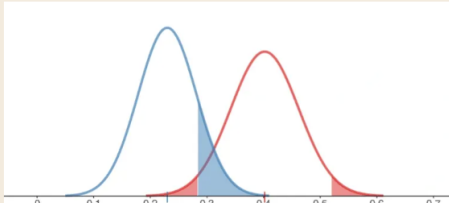
Find **confidence intervals** or test **hypotheses** about a population *mean*. Enter your own data or summary statistics. Use plots to check assumptions and visualize the interval or the **P-value** on a graph.

Create and visualize, step-by-step, the **bootstrap distribution** of the *mean*, *median* or *standard deviation*. Obtain summary statistics and find a percentile confidence interval.

Explore Coverage



Errors and Power



Art of Stat: Inference





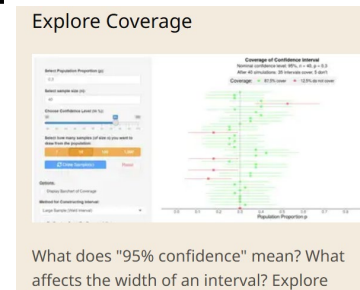
Your Turn: Our In-class Lab for this Chapter

(a) Use the Explore Coverage Web App at www.artofstat.com/web-apps, to explore the performance of confidence intervals for a proportion when n is not very large and π may be near 0 or 1, by repeatedly generating random samples and constructing the confidence intervals. Take the population proportion $\pi = 0.06$, $n = 20$, and confidence level 95%. What proportion of generated confidence intervals actually captured the true value of the parameter?

(b) Use the Wald confidence interval (formula (4.8)). Choose 10 samples of size $n = 20$. Now draw another 10. How many of the 20 Wald intervals contained the true value, $\pi = 0.06$? How many would you expect to contain it? To see that this is not a fluke, simulate by drawing 1000 random samples of size $n = 20$ and constructing 1000 confidence intervals. What percentage contain $\pi = 0.06$? What does this suggest?

(c) Using the *Sampling Distribution of the Sample Proportion* app at www.artofstat.com/web-apps, let $\pi = 0.06$ and draw 10,000 random samples of size 20 each. Look at the simulated sampling distribution of $\hat{\pi}$, which is highly discrete. Is it bell-shaped and symmetric? Use this to help explain why the Wald confidence interval performs poorly in this case.

(d) Now use the Wilson's score interval (formula (4.7)) within the *Explore Coverage* app. Draw 1000 samples of size 20 each when $\pi = 0.06$. What percentage of the 95% confidence intervals actually contained π ? How does this compare to the value from (b)?



Method for Constructing Interval:

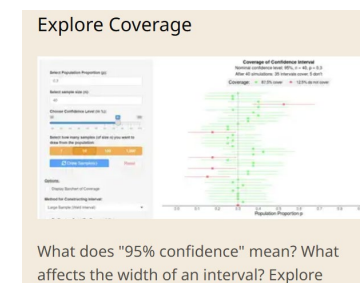
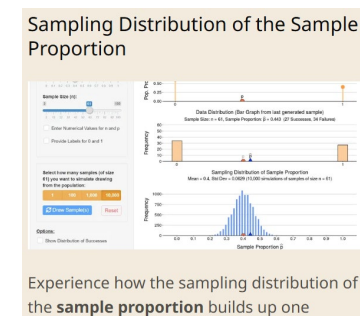
Large Sample (Wald Interval) ▼

Large Sample (Wald Interval)

Agresti-Coull

Wilson Score

Exact (Clopper-Pearson)



Lab Continued.

After you complete the lab activities, discuss the following, explain what is wrong with each of the provided explanations based on a scenario in which a random sample of 50 records yields a 95% confidence interval for the mean age at first marriage of women in a certain county in Virginia was 23.5 to 25.0 years.

A. If random samples of 50 records were repeatedly selected, then 95% of the time the sample mean age at first marriage for women would be between 23.5 and 25.0 years.

B. Ninety-five percent of the ages at first marriage for women in the county are between 23.5 and 25.0 years.

C. We can be 95% confident that $\bar{y}$ is between 23.5 and 25.0 years.

D. If we repeatedly sampled the entire population, then 95% of the time the population mean would be between 23.5 and 25.0 years.